# Draft appendix: uncertainty of the network model

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## A Uncertainty of the network model

The differential cross section is modelled as $\sigma(m_{hh}, \theta) = \sum_{i=1}^{23} c_i(\theta)\, a_i(m_{hh})$, where the monomials $c_i(\theta)$ in the HEFT parameters $\theta = (\kappa_\lambda, c_t, c_{tt}, c_{ggh}, c_{gghh})$ are known exactly [1] and the 23 functions $a_i(m_{hh})$ are represented by a neural network (section **??**). In this appendix we describe how the network is trained, how its uncertainty is estimated, and how both are validated on POWHEG spectra not used in the training.

### A.1 Data and noise model

The training sample consists of 3,989 POWHEG spectra at 13 TeV, each at a different point $\theta$, binned in 80 bins of 15 GeV between $m_{hh} = 255$ and 1455 GeV. Each bin of each spectrum is one training row, $(m_{hh}, \theta) \mapsto \sigma$. Of the 317,306 rows, 274,152 are used for training, 5,078 for validation (the choice of the saved parameters), and 38,076 are not used in the training. A further 1,000 spectra, generated in the same way, are held out completely and serve as the test sample.

Every spectrum is, to a good approximation, a histogram of $N_{\text{eff}}$ events of equal weight. We estimate $N_{\text{eff}}$ from the bin-to-bin fluctuations of the training spectra in the smooth region $m_{hh} > 570$ GeV, comparing each bin with the average of its two neighbours, and find $N_{\text{eff}} \simeq 3.8 \times 10^4$. The noise variance of bin $k$ of a spectrum with total cross section $\sigma_{\text{tot}}$ is then taken to be

$$\sigma_k^2 = \Big(t_k + \frac{\sigma_{\text{tot}}}{N_{\text{eff}}}\Big)\frac{\sigma_{\text{tot}}}{N_{\text{eff}}}, \tag{1}$$

where $t_k$ is the POWHEG cross section in the bin; the second term, the variance of one additional event, prevents bins with very few events from acquiring unphysically small variances. For the test sample we use $\sigma_k^2 = t_k \sigma_{\text{tot}}/N_{\text{eff}}$.

### A.2 Training

Each network is trained in two stages.

**Stage 1.** The network is trained by minimising the mean absolute deviation (L1 loss) between prediction and POWHEG, with the Adam optimiser [2], mini-batches of 2,048 rows, $2 \times 10^5$ iterations and a learning rate of $2 \times 10^{-3}$ halved three times at equal intervals. The parameters with the lowest validation loss are kept. The L1 loss treats an absolute error equally in every bin, so the fit is dominated by the largest cross sections, and the region just above threshold, where the cross section is two orders of magnitude below its peak, is modelled less well (table 1).

**Stage 2.** The network is then fine-tuned with the Huber loss [3] of the pulls $r_k = (f_k - t_k)/\sigma_k$, where $f_k$ is the network prediction,

$$\ell_\delta(r) = \begin{cases} \tfrac{1}{2} r^2, & |r| \le \delta, \\ \delta\big(|r| - \tfrac{1}{2}\delta\big), & |r| > \delta, \end{cases} \tag{2}$$

with $\delta = 2$. For bins within two standard deviations of POWHEG this is $\chi^2/2$, so each bin is weighted by its precision; bins further away contribute linearly, which limits the influence of the few badly modelled bins and of spectra whose noise is underestimated. The fine-tune uses gradients computed on all training rows at every step ("full batch"), Adam with a learning rate of $10^{-4}$ halved three times, and $10^4$ steps; the parameters with the lowest validation loss (of the same Huber loss) are kept. Starting from the stage-1 solution is essential: trained from random weights with mini-batches, a $\chi^2$ loss converges poorly, presumably because the weights $1/\sigma_k^2$ span many orders of magnitude. On a single network, the results depend little on $\delta$: for $\delta = 1$, 2, 3 and $\infty$ (pure $\chi^2$), the mean $\chi^2$ per bin between 255 and 555 GeV is 1.66, 1.63, 1.63 and 1.59, against 8.44 after stage 1.

### A.3 Ensembles

We regard the training procedure as an estimator $\hat{f}(\boldsymbol{x}; D, s)$ of the cross section at $\boldsymbol{x} = (m_{hh}, \theta)$, which depends on the training sample $D$ and on a random seed $s$ that fixes the initial weights and the order of the mini-batches. Its standard deviation at fixed $\boldsymbol{x}$ is the standard error of the network prediction. We estimate it with two ensembles of $M = 20$ networks each:

- a *seeds* ensemble: the same training rows, seeds $s = 42, \ldots, 61$;
- a *bootstrap* ensemble [4, 5]: member $m$ is trained, with seed $42 + m$, on 274,152 rows drawn with replacement from the training rows (on average 63% distinct).

The ensemble mean $\bar{f}$ is the prediction, and the standard deviation $s$ of the members its uncertainty; with $M = 20$, $s$ is itself uncertain by about 16%. The bootstrap treats the training rows as independent draws from a population of rows and estimates the variability of the result under a repetition of the whole procedure with a new sample of the same size. This is one of several defensible reference ensembles: one could instead resample whole spectra (new points $\theta$) or regenerate the POWHEG events at fixed $\theta$; the choice is not dictated by the data [6, 7]. Because member $m$ uses the same seed in both ensembles, the law of total variance,

$$\mathrm{Var}_{D,s}[\hat{f}] = \mathrm{E}_D\left[\,\mathrm{Var}_s[\hat{f} \mid D]\right] + \mathrm{Var}_D\left[\,\mathrm{E}_s[\hat{f} \mid D]\right], \tag{3}$$

separates the bootstrap variance (left) into an optimisation term, estimated by the seeds ensemble, and a data term, their difference. Ensembles of independently trained networks are known to capture the variability between equally good solutions ("deep ensembles") [8, 9].

### A.4 Bayesian cross-check

As a cross-check we sampled the posterior density of the 3,944 network parameters following Neal [10], with Hamiltonian Monte Carlo [11, 12], using a hierarchical Gaussian prior with one Gibbs-sampled precision per parameter tensor and a Gaussian likelihood with the known variances (1). The step size and a diagonal mass matrix were adapted during a warm-up of $10^3$ iterations [13], after which $10^3$ samples were drawn with 800 leapfrog steps per trajectory, starting

| | median $\chi^2/N$ | $\chi^2/N > 5$ | $\langle\chi^2\rangle$ per bin | | negative |
|---|---|---|---|---|---|
| | | | 255–555 GeV | $> 555$ GeV | predictions |
| single network, stage 1 | 1.171 | 4.9% | 8.44 | 1.20 | 0.18% |
| stage-1 ensemble, seeds | 1.127 | 2.3% | 4.08 | 1.17 | 0.06% |
| stage-1 ensemble, bootstrap | 1.118 | 2.2% | 4.05 | 1.14 | 0.06% |
| fine-tuned ensemble, seeds | 1.068 | 0.4% | 1.62 | 1.10 | 0.004% |
| fine-tuned ensemble, bootstrap | 1.066 | 0.4% | 1.62 | 1.10 | 0.004% |
| Bayesian network (posterior mean) | 1.099 | 1.2% | 1.89 | 1.12 | 0.03% |

Table 1: Accuracy on the 1,000 test spectra. $\chi^2/N$ is computed per spectrum; the columns give its median and the fraction of spectra above 5, and the mean $\chi^2$ per bin below and above 555 GeV. For the ensembles, the mean of the 20 members is used.

from the stage-1 network. The effective sample size of the predictions at the monitored points is about 135. The posterior mean is close in accuracy to the fine-tuned ensemble (table 1) and differs from the ensemble mean by a median of 1.1 ensemble standard deviations. The posterior standard deviation, however, is only 0.5% of the POWHEG noise: with $2.7 \times 10^5$ training rows the parameters of a single network are tightly constrained, and the posterior describes neither the variability between equally good networks, which a single Markov chain does not explore, nor the model's own error. We therefore do not use it as the uncertainty.

### A.5 Validation on the test spectra

On the 1,000 test spectra we compute, for every bin, the $\chi^2$ contribution $(t_k - \bar{f}_k)^2/\sigma_k^2$ and the pull

$$z_k = \frac{t_k - \bar{f}_k}{\sqrt{\sigma_k^2 + s_k^2}}. \tag{4}$$

If $s_k$ accounts for the difference between the network and the true cross section, $z$ has unit standard deviation. Table 1 and figure 1 show the accuracy, table 2 and figure 2 the calibration, and figure 3 the decomposition (3).

**Accuracy.** Averaging over an ensemble improves on a single network, and the fine-tune improves substantially on stage 1, above all near threshold, where the mean $\chi^2$ per bin falls from 8.4 (single network) and 4.1 (stage-1 ensemble) to 1.6. Negative predictions, which are unphysical, become very rare. The fine-tuned members agree closely with one another (their $\chi^2$ per bin near threshold ranges from 1.6 to 1.8, against 3.6 to 15 for the stage-1 members).

**Calibration.** The spread of the stage-1 ensembles matches their errors: the pull has unit standard deviation within 2–3% everywhere. Their members, however, disagree strongly near threshold, which is also where their mean is least accurate. The fine-tune removes most of that disagreement and most of the error, but the spread shrinks by a factor of three, more than the error does: what remains is an error common to all members, which no ensemble of identically trained networks can reveal. The pull of the fine-tuned ensemble has a standard deviation of 1.15 near threshold and 1.04 above. The Bayesian posterior is far too narrow, as explained in appendix A.4. The tails of the pull distribution are somewhat heavier than Gaussian for all methods: for the stage-1 ensembles, 0.56% of bins have $|z| > 3$, against 0.27% for a standard normal.

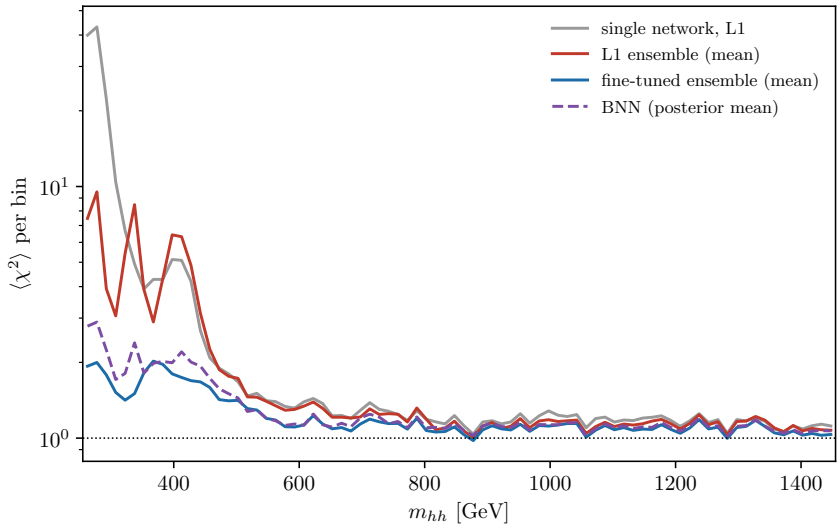



Figure 1: Mean $\chi^2$ per bin over the 1,000 test spectra as a function of $m_{hh}$, for a single stage-1 network, the stage-1 (seeds) ensemble, the fine-tuned (bootstrap) ensemble and the posterior mean of the Bayesian network.

| | median $s/\sigma_k$ | | standard deviation of $z$ | | | $\lvert z\rvert < 2$ |
|---|---|---|---|---|---|---|
| | all | 255–555 GeV | all | 255–555 GeV | $> 555$ GeV | |
| stage-1 ensemble, seeds | 0.157 | 0.25 | 1.024 | 1.018 | 1.026 | 95.0% |
| stage-1 ensemble, bootstrap | 0.157 | 0.25 | 1.025 | 1.020 | 1.026 | 95.0% |
| fine-tuned ensemble, seeds | 0.051 | 0.075 | 1.076 | 1.170 | 1.042 | 94.2% |
| fine-tuned ensemble, bootstrap | 0.056 | 0.091 | 1.069 | 1.145 | 1.042 | 94.2% |
| with model error, eq. (6) † | 0.32 | 0.47 | 1.003 | 1.024 | 0.995 | 95.6% |
| Bayesian network | 0.005 | 0.007 | 1.143 | 1.370 | 1.055 | 93.5% |

Table 2: Calibration on the test spectra: the uncertainty $s$ in units of the POWHEG noise, and the pull (4). For a standard normal pull, $|z| < 2$ holds for 95.4% of bins. †Here the first two columns give the total uncertainty $s_{k,\mathrm{tot}}/\sigma_k$ of eq. (5), with $a$ determined on training-file rows not used in the training (appendix A.6).

**Sources of the spread.** For the stage-1 networks the bootstrap and seeds variances are equal within their statistical uncertainty (median ratio 1.05): the spread is entirely due to the optimisation, i.e. to the different solutions reached from different starting points, and resampling the training rows adds nothing measurable. For the fine-tuned networks the optimisation term accounts for about 80% of the variance and the data term for about 20%, corresponding to standard deviations of about 5% and 2% of the POWHEG noise. With $2.7 \times 10^5$ training rows, the statistical uncertainty due to the finite training sample is thus small compared with the POWHEG noise. The means of the seeds and bootstrap fine-tuned ensembles agree to 0.2 of their spread.

## A.6 Model-error term

To account for the error common to all members, we add a model-error term in quadrature,

$$s^2_{k,\mathrm{tot}} = s^2_k + a(m_{hh})\,\sigma^2_k, \tag{5}$$

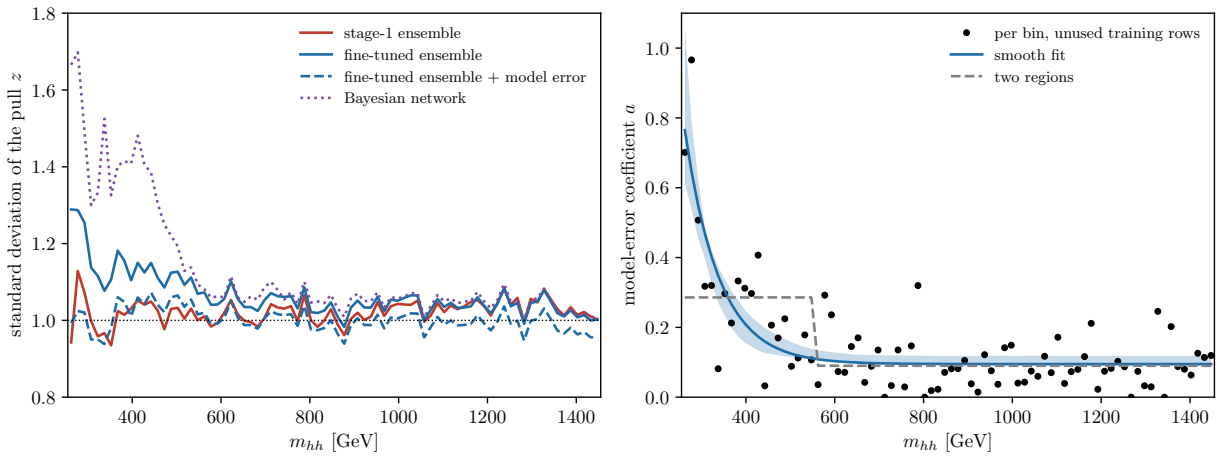



Figure 2: Left: standard deviation of the pull (4) on the test spectra as a function of $m_{hh}$; the dashed curve includes the model-error term (5). Right: the model-error coefficient $a$ determined bin by bin on the 38,076 training-file rows not used in the training (points), the fit (6) with its 68% band from a bootstrap over spectra (solid), and the two-region estimate (dashed).

and replace $s_k^2$ by $s_{k,\mathrm{tot}}^2$ in the pull (4). The coefficient is determined on the 38,076 rows of the training file that were used neither for training nor for validation, so that the test spectra serve only for validation. For each of the 80 bins we solve $\langle z^2 \rangle = 1$ for $a$; the results (figure 2, right) are large just above threshold and roughly constant beyond about 500 GeV, and are well described by

$$a(m_{hh}) = a_0\, e^{-(m_{hh}-255\,\mathrm{GeV})/\lambda} + a_1, \qquad a_0 = 0.74, \quad \lambda = 77\ \mathrm{GeV}, \quad a_1 = 0.095, \tag{6}$$

fitted by least squares; a bootstrap over the spectra gives uncertainties of 0.28, 35 GeV and 0.013 on the three parameters, which are strongly correlated. A cruder estimate with one coefficient below and one above 555 GeV gives $a = 0.29 \pm 0.03$ and $0.090 \pm 0.013$. The model error is thus about 0.9 of the POWHEG noise in the first bin, 0.5 at 400 GeV and 0.3 above 600 GeV, several times the spread of the ensemble.

On the 1,000 test spectra, with $a$ from eq. (6), the pull has a standard deviation of 1.003 (1.024 below and 0.995 above 555 GeV), and $|z| < 1$ and $|z| < 2$ hold for 69.3% and 95.6% of the bins, against 68.3% and 95.4% for a standard normal (table 2, figure 2). The two-region estimate performs almost as well (1.003; 1.018 and 0.998). The same coefficients also calibrate the seeds ensemble (1.008).

Two remarks. First, the residuals have heavier tails than a Gaussian: requiring instead that the robust standard deviation $1.4826 \times \mathrm{median}|z|$ be one gives $a = 0.13$ below 555 GeV, half the value obtained from the variance, so near threshold the bulk of the bins is described better than the variance suggests, and a few bins are described worse. Second, above 555 GeV the term $a_1\sigma_k^2$ cannot be distinguished from a 10% underestimate of the POWHEG variance, i.e. of $1/N_{\mathrm{eff}}$ in eq. (1); the procedure calibrates the total uncertainty but does not tell the two apart.

## A.7 Summary

The cross-section model used in this paper is the mean of an ensemble of 20 networks, each trained with an L1 loss and fine-tuned with a Huber loss of the pulls, and its uncertainty is the standard deviation of the members combined in quadrature with a model-error term of 0.3–0.9 of the

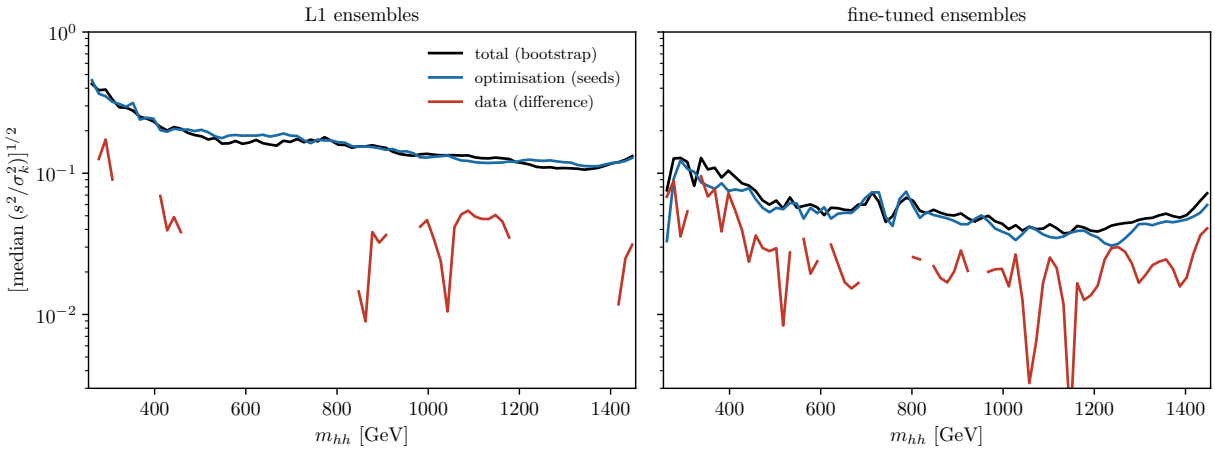



Figure 3: Decomposition (3) of the ensemble variance, for stage-1 (left) and fine-tuned (right) ensembles: total (bootstrap), optimisation (seeds) and their difference (data), as the square root of the median over the test spectra of $s^2/\sigma_k^2$. Where the seeds variance exceeds the bootstrap variance, which happens within the statistical uncertainty of 20-member ensembles, the data term is not shown.

POWHEG noise, determined on training-file rows not used in the training. On 1,000 independent POWHEG spectra this model reaches a median $\chi^2/N$ of 1.07, and its uncertainty is calibrated to within 2.5% at all $m_{hh}$. The uncertainty is dominated by the model-error term; the spread of the ensemble, about 80% of which reflects the variability between equally good networks and 20% the finite training sample, is several times smaller. Both are largest just above threshold. A Bayesian treatment of the network parameters gives a compatible central value, but its posterior width is negligible and does not describe the uncertainty of the model.

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